Medical Diagnosis and Report Generation for Endoscopic Procedures Using Vision-Language Models

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2025

CERTIFICATE

Certified that this is a bonafide record of the project work titled

MEDICAL DIAGNOSIS AND REPORT GENERATION FOR ENDOSCOPIC PROCEDURES USING VISION-LANGUAGE MODELS

done by

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of eighth semester B. Tech in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of the National Institute of Technology Calicut

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DECLARATION

We hereby declare that the project titled, **Medical Diagnosis and Report Generation for Endoscopic Procedures Using Vision-Language Models**, is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or any other institute of higher learning, except where due acknowledgement and reference has been made in the text.

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Abstract

Background and Objective: Automated generation of endoscopic examination reports is a clinically significant yet technically demanding problem. Producing such reports demands deep familiarity with domain-specific terminology, pathological presentation, and the structured reasoning conventions of clinical practice, a burden that consumes considerable time even for experienced endoscopists. While multimodal large language models (MLLMs) have demonstrated impressive capabilities in joint visual and language processing, their utility in specialised medical domains is constrained by inadequate command of clinical vocabulary, difficulty recognising subtle pathological features, and limited adherence to medical reporting logic. Standard fine-tuning approaches partially address these gaps, yet commonly fail to provide the depth of domain-specific adaptation needed for high-quality endoscopic report generation. Methods: Building on the two-stage prompt-based LoRA (P-LoRA) framework introduced by Zhang et al. [42], we develop MVLMERG (Multimodal Vision-Language Model for Endoscopic Report Generation). The first stage enriches the base model with broad medical domain knowledge sourced from publicly available datasets, while the second stage specialises the model for clinical endoscopic reporting through fine-tuning on physician-annotated data. Results: On our curated endoscopic dataset, MVLMERG achieves state-of-the-art results, surpassing all evaluated baseline systems across every automated evaluation metric.

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Chapter 1

Introduction

Project Information

Title: Medical Diagnosis and Report Generation for Endoscopic Procedures Using Vision-Language Models.

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1.1 Introduction

Endoscopic examinations occupy a central role in the early identification, diagnosis, and clinical management of gastrointestinal conditions [3,4], and a growing body of work has continually advanced the state of the art for associated computational tasks [1,2]. Every procedure produces substantial volumes of visual data that clinicians must interpret meticulously before composing accurate written reports, a process that is simultaneously cognitively demanding and time-intensive. Multimodal large language models (MLLMs) offer a compelling path forward: by pre-training on broad-scale corpora, they acquire transferable general knowledge that can subsequently be directed toward narrow specialised domains, making them well-suited to bridging the gap between raw endoscopic imagery and structured diagnostic text [5].

Constructing unified multimodal systems for clinical practice has led researchers to explore architectures capable of integrated cross-modal reasoning [6,7]. Large biomedical repositories such as PubMed [8] furnish extensive general medical knowledge, yet their coverage of pathology-specific or organ-level annotations remains limited. The scarcity of carefully labelled endoscopic image data further constrains the development of robust diagnostic systems. Compounding these data-availability challenges, building large multimodal architectures from the ground up demands prohibitive computational investment; consequently, parameter-efficient adaptation of pre-existing models has emerged as a far more tractable strategy [9].

Although LoRA demonstrates broad transferability across diverse tasks, its adaptation performance on highly specialised downstream problems can fall short of clinical requirements [11]. Combining prompt-based conditioning mechanisms with LoRA inside the LLaVA-4 framework therefore constitutes a pressing open challenge for domain-specific deployment.

Following Zhang et al. [42], whose staged fine-tuning methodology directly addresses these limitations, this project implements MVLMERG (Multimodal Vision-Language Model for Endoscopic Report Generation) and demonstrates that it achieves state-of-the-art performance relative to mainstream foundational models. The training corpus is built from physician-annotated endoscopic records subsequently standardised with GPT-4. Adopting the two-stage P-LoRA framework of [12,42], MVLMERG first undergoes broad medical domain enrichment that equips the model with the clinical vocabulary and conceptual grounding required to reason accurately about endoscopic findings. A second fine-tuning stage then steers the model toward producing outputs that conform to the structural and stylistic conventions of professional endoscopic documentation.

The principal contributions of this work are as follows:

1. A specialised vision-language system, MVLMERG, purpose-built for clinical endoscopic report generation, which surpasses GPT-4V, HuatuoGPT-Vision-7B, and other competitive baselines across every automated evaluation metric on our private endoscopic dataset.

2. A two-stage prompt-guided LoRA (P-LoRA) fine-tuning strategy that explicitly separates the acquisition of medical domain knowledge from the learning of professional report-generation conventions, enabling effective adaptation of LLaVA-4 to the endoscopic reporting task with substantially reduced computational overhead.

The remainder of this report proceeds as follows. Chapter 2 surveys the relevant literature on medical documentation burden, multimodal large models, and parameter-efficient fine-tuning. Chapter 3 formally defines the problem addressed by this work. Chapter 4 presents the MVLMERG architecture and methodology in full, covering the two-stage fine-tuning pipeline, the P-LoRA mechanism, dataset construction and preprocessing, image augmentation procedures, and chain-of-thought reasoning integration. Chapter 5 reports experimental findings and quantitative comparisons against baseline systems. Chapter 6 draws conclusions and outlines directions for future research.

Chapter 2

Literature Survey

2.1 Evolution of Medical Documentation Burden

Over recent decades, administrative documentation tasks have come to occupy a growing share of clinical working hours, with primary care physicians bearing a disproportionate part of this burden [40]. The 2019 National Electronic Health Records Survey provides concrete measurements: physicians spend an average of 1.77 hours per day on documentation activities conducted outside of scheduled office time, and this figure rises to 1.84 hours among those who use EHR systems, against only 1.10 hours for non-EHR users [41].

The effects reach further than time loss alone. Surveys indicate that 75% of healthcare providers consider current documentation demands to be actively harmful to the quality of care they can deliver [40]. At a more granular level, 58.1% of physicians reject the view that their documentation workload is appropriately sized or that it preserves sufficient time for patient interaction, and 84.7% acknowledge that record-keeping driven purely by billing obligations inflates their total documentation hours [40,41]. Collectively, these figures make a compelling case for intelligent automation tools that can absorb documentation overhead while maintaining the accuracy and professional standards required in clinical records.

2.2 MLLMs

Large-scale pretraining has given rise to multimodal large language models (MLLMs) that jointly process visual and textual information, dramatically broadening the scope of AI perception and generation. A central design challenge in this area is achieving deep, rather than superficial, coupling between vision and language components. Wang et al. [13] tackled this problem through CogVLM, which inserts trainable visual expert modules directly into the attention and FFN layers of a frozen language backbone, enabling parameter-level fusion of visual and linguistic representations without degrading natural language processing behaviour. Approaching the same challenge from a different angle, Bai et al. [14] introduced the Qwen-VL model family to remedy the difficulty that standard language models face when interpreting raw visual signals; a three-stage curriculum training procedure applied to multilingual data allowed Qwen-VL to set new performance marks on image captioning, visual question answering, and visual grounding evaluations. Liu et al. [15] pursued yet another strategy, exploiting GPT-4's text-only reasoning capacity to synthesise multimodal instruction-following data, yielding LLaVA, a system that couples a vision encoder with a large language model backbone and achieves comparable results to multimodal GPT-4 on images unseen during training. Despite these advances, clinical data constitute only a small fraction of the corpora used to train such systems, leaving general-purpose models poorly equipped to characterise lesion morphology accurately or to produce outputs that conform to the structured conventions of professional diagnostic reporting.

2.3 Medical MLLMs

Parallel to the growth of general-purpose vision-language research, the healthcare domain has steadily accumulated heterogeneous multimodal data (spanning clinical images, operative recordings, histological slides, structured laboratory results, and longitudinal health records) that now underpin a new generation of medically specialised MLLMs. Concerning the quality and breadth of medical image–text supervision, Chen et al. [16] constructed the PubMedVision dataset by harvesting image–text pairs from PubMed and subjecting them to an MLLM-assisted denoising step; training HuatuoGPT-Vision on this curated resource produced marked gains across a range of medical multimodal benchmarks. In surgical settings, Seenivasan et al. [17] observed that the unidirectional autoregressive nature of GPT-style decoders prevents direct ingestion of surgical image inputs, and responded with SurgicalGPT, an end-to-end trainable architecture that resolves this constraint and achieves accuracy improvements of 3%–5% across three surgical evaluation sets. For colonoscopy in particular, Ji et al. [18] identified a gap in both task-appropriate benchmarks and existing model capability, addressing it by releasing the ColonINST benchmark alongside the ColonGPT model, which surpassed prior approaches on colonoscopy classification, referring expression, and localisation while demonstrating stronger cross-task generalisation. Targeting the broader problem of narrow task coverage and limited transferability that affects most surgical vision-language systems, Schmidgall et al. [19] released GP-VLS, an open-source general-purpose model that delivered consistent accuracy gains across a diverse suite of surgical vision-language evaluations.

2.4 Fine-tuning methods

The prohibitive cost of pretraining large multimodal models from scratch, combined with the performance degradation that results from deploying unadapted general models on narrow clinical tasks, has established task-specific fine-tuning as the dominant adaptation strategy for medical MLLMs [20,21]. Li et al. [7] provide a prominent illustration with LLaVA-Med, which approaches biomedical image understanding via a two-stage pipeline: an initial alignment phase on the PMC-15M corpus anchors the model in biomedical vocabulary, after which GPT-4-generated instruction data drives full end-to-end instruction tuning. Concerned with the parameter overhead that medical MLLM adaptation typically incurs, He et al. [22] proposed PeFoMed, a framework whose two-stage fine-tuning design achieves competitive results on medical imaging benchmarks without proportionally expanding the set of trainable parameters. Chen et al. [23] focused on the twin obstacles of data privacy restrictions and scarce expert labels that frequently hinder medical visual question answering, introducing a generative pretraining and fine-tuning pipeline designed to operate effectively under these constraints. Lu et al. [24] demonstrated that casting visual features as soft visual prompts aligned with a language model's text embedding space, combined with a two-stage fine-tuning schedule, yields meaningful improvements in both the accuracy and generation efficiency of radiology report synthesis. In pathology, Wu et al. [25] assembled PathEnhanceDS, a 45,000-case instruction-tuning corpus, and showed that it raises model competence on pathological image classification, descriptive captioning, and question answering tasks. Notwithstanding these contributions, prevailing fine-tuning approaches share three persistent shortcomings: (1) clinical domain knowledge is insufficiently embedded during the fine-tuning process, leaving models with shallow comprehension of specialised medical semantics; (2) existing systems lack flexible, task-aware prompting mechanisms that can steer model attention toward the most diagnostically salient visual features; and (3) the practice of applying LoRA uniformly across every transformer layer is computationally wasteful, since it ignores the functionally distinct roles that early, middle, and late layers each play in multimodal representation learning.

Chapter 3

Problem Definition

Automated endoscopic report generation sits at the intersection of computer vision and clinical natural language generation, and presents challenges that general-purpose multimodal models are ill-equipped to handle. Physicians who interpret endoscopic examinations must integrate visual findings with specialised clinical knowledge and produce structured reports conforming to established medical conventions, a process that is both cognitively demanding and time-consuming. General-purpose MLLMs, despite their broad cross-modal capabilities, frequently misidentify pathological features, misuse domain terminology, and produce outputs that lack the structured reasoning required in endoscopic practice. Following Zhang et al. [42], who identified these limitations and proposed a staged fine-tuning strategy to address them, this project adopts a similar problem framing: effective endoscopic report generation requires not merely task-specific fine-tuning but a deliberate injection of medical domain knowledge followed by alignment to professional reporting conventions. The objective of this project is therefore to design and implement MVLMERG, a system capable of bridging the knowledge gap of general vision-language models, incorporating clinically relevant domain understanding, and generating accurate, structured endoscopic reports that meet professional medical standards.

Chapter 4

Methodology

4.1 Method

In this report, we propose a two-stage P-LoRA method [42] and fine-tune LLaVA-4 as the base model, effectively improving its ability to describe endoscopic images.

4.1.1 Overview

Fig. 1 depicts the overall architecture employed for fine-tuning a multimodal large language model through P-LoRA. Training proceeds across two sequential stages. In Stage 1, a broad corpus of publicly available endoscopy-related material—including diagnostic reports, prescriptions, clinical dialogues, and analogous content—is assembled as the primary source of domain knowledge. Following Zhang et al. [42], P-LoRA weight updates are then applied to transfer this knowledge into the language model while the visual encoder and connector parameters remain frozen throughout. Stage 2 is conducted in collaboration with expert physicians at Government Medical College, Calicut, who contributed a private multimodal dataset comprising gastric endoscopic images paired with annotated textual descriptions. In this stage, the visual encoder and connector continue to be held fixed; parameter updates are concentrated on the language model alone, equipping the system to produce clinically grounded endoscopic reports. This staged training regime, as proposed by Zhang et al. [42], reduces computational overhead while enabling the model to progressively integrate visual and linguistic modalities for accurate report synthesis. The open-source MLLM LLaVA-4 [39] serves as the backbone of our framework. Visual features are derived from gastric endoscopic images using the pretrained CLIP-ViT-L/14 encoder [15,39]. Given an input RGB image $I \in \mathbb{R}^{H \times W \times 3}$, where H and W denote the image height and width respectively, the visual encoder produces a visual token sequence $V = [v_1, v_2, \ldots, v_{N_v}]$, with N_v being the total number of visual tokens. This sequence is forwarded to the Vicuna-7B-v1.5 language model backbone. The visual tokens are then concatenated with the text token sequence $T = [t_1, t_2, \ldots, t_{N_t}]$, where N_t denotes the number of text tokens, and the unified sequence is processed by the multimodal language model to produce a language response $k = [r_1, r_2, \ldots, r_{N_r}]$, where N_r is the response length.

4.1.2 Two-stage fine-tuning strategy

Standard MLLMs lack sufficient exposure to endoscopy-specific knowledge, leaving significant gaps in their capacity to interpret lesion morphology and gastroenterological terminology. Following Zhang et al. [42], we address this deficit through a two-stage fine-tuning strategy that decouples domain knowledge acquisition from clinical report generation. This design allows the model to first build a broad medical foundation before being specialised toward the stylistic and structural standards of professional endoscopic documentation, yielding outputs that are both medically accurate and clinically consistent. Stage 1 draws on three complementary publicly available datasets that collectively supply the gastroenterological breadth required for downstream specialisation.

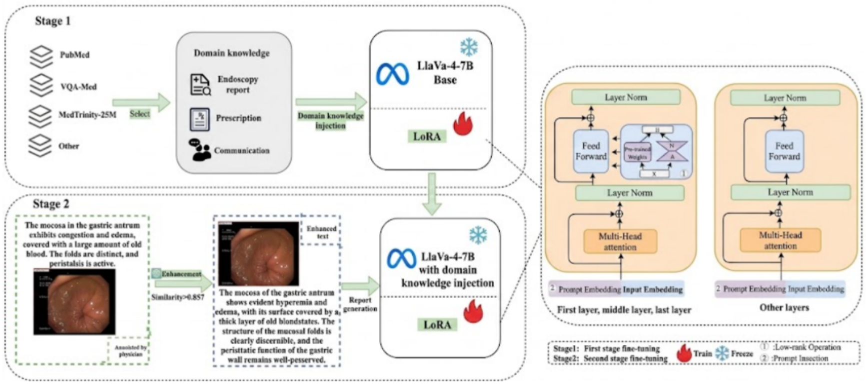


Figure 4.1: Overall architecture of the proposed two-stage fine-tuning pipeline. The left portion depicts Stage 1 (public-dataset domain injection) and Stage 2 (private clinical fine-tuning); the right portion provides a detailed view of the P-LoRA adaptation module. Green-highlighted components indicate the elements introduced in this work.

PubMed [8] indexes more than 35 million biomedical citations and abstracts from life science journals worldwide. Our training material is sourced from the PubMed Central (PMC) Open Access subset, which supplies full-text articles together with embedded figure–caption pairs. A curated gastroenterology vocabulary is applied to retain only GI-relevant image–text records spanning the esophagus, stomach, duodenum, and colon. This filtered subset establishes semantically rich grounding in anatomical expressions, lesion terminology, and clinical descriptors that form the foundation for multimodal pretraining.

MedTrinity-25M [26] is a large-scale collection of approximately 25 million image–text pairs organised as (image, region, enriched description) triplets across diverse clinical modalities, including radiology (X-ray, CT, MRI), histopathology slides, and endoscopy. Each triplet associates a localised image region with a detailed, grounded textual description, supporting fine-grained visual–language alignment. The clinical breadth and scale of this dataset make it particularly suited to exposing the model to diverse pathological appearances and their corresponding diagnostic vocabulary.

VQA-Med [27], derived from the ImageCLEF 2019 medical VQA challenge, provides over 4,200 radiology images each paired with targeted clinical question–answer items spanning four question types: modality recognition, plane identification, organ identification, and abnormality detection. Exposure to this dataset develops the model's capacity to associate image-level visual evidence with structured clinical reasoning, strengthening its response accuracy on domain-specific queries and complementing the descriptive knowledge furnished by PubMed and MedTrinity-25M.

Together, these three datasets supply foundational knowledge of gastric anatomy, lesion localisation, tissue-level structural changes, and standard diagnostic terminology. Throughout this stage, vision encoder and connector weights are frozen, and parameter updates are restricted to the language model alone. In Stage 2, physician-annotated records from Government Medical College, Calicut are used to align the model's language generation with real-world clinical reporting practice. The data undergo the targeted processing pipeline described in Section 4.1.4 before fine-tuning. This stage prioritises accurate command of professional terminology, discipline-consistent sentence structure, and the specific expression patterns characteristic of formal endoscopic reports, producing outputs that faithfully reproduce the documentation standards expected in clinical settings.

4.1.3 P-LoRA method

While LoRA achieves broad generalisation across tasks, its adaptation performance on narrow domain-specific problems can be insufficient when applied in isolation [11]. As proposed by Zhang et al. [42], our P-LoRA method addresses this limitation by coupling learnable soft prompt tokens with layer-selective LoRA weight updates within a unified parameter-efficient framework. Soft prompts are continuous, task-optimised embedding vectors prepended to the input sequence; unlike hard prompts, which encode manually crafted discrete tokens, soft prompts reside directly in the embedding space and are refined end-to-end through gradient descent with no manual engineering required. This mechanism guides the model toward domain-appropriate representations at minimal parameter cost. Selective LoRA weight increments then provide complementary, targeted adjustments that sharpen feature extraction across both visual and textual streams. Neither component alone is sufficient: soft prompts without weight updates may fail to reshape deep internal representations, whereas unrestricted LoRA incurs higher parameter overhead. Following Zhang et al. [42], LoRA is restricted to three structural positions in the network—the first layer, the middle layers, and the final layer—reflecting the empirically observed principle that shallow layers encode general low-level features, intermediate layers mediate cross-modal interactions, and deep layers specialise for task-specific output, as illustrated in Fig. 2. As shown in Fig. 2(a), trainable prompt parameters [28] are inserted at every attention layer, with the soft prompt tokens initialised from the same normal distribution as the pretrained embedding layer. Formally, a set of learnable prompt tokens $p = (p_1, p_2, \ldots, p_m)$ is prepended to the original input embedding sequence $x = (x_1, x_2, \ldots, x_n)$, where each $x_i \in \mathbb{R}^{d_{model}}$ and d_{model} is the model's hidden dimensionality. The resulting extended input $x' = (p_1, p_2, \ldots, p_m, x_1, x_2, \ldots, x_n)$ is forwarded to the attention layer, where queries, keys, and values are computed over the full extended sequence. Domain-relevant signals encoded in the prompt tokens thereby influence the attention computation across all positions without altering the pretrained backbone weights, and the output dimensionality remains equal to d_{model}. The prompt tokens adapt entirely through task-driven gradient updates; no manually crafted content is introduced. Low-rank weight updates via LoRA are subsequently applied to the feed-forward layers, as depicted in Fig. 2(b). Restricting LoRA to the first, middle, and final Transformer layers serves distinct functional purposes: the first layer is adapted to realign the model's sensitivity to novel domain-specific input patterns; the middle layers are updated to recalibrate attention over gastroenterological features; and the final layer is tuned to condition output generation on task-specific requirements. Within each selected feed-forward block, the first linear projection expands the representation from d_{model} to d_{ff}, and a low-rank increment is introduced at this weight matrix. Rather than updating the full weight matrix $W \in \mathbb{R}^{m \times n}$ as in conventional fine-tuning, the pretrained weights are frozen and only an additive term ΔW is learned, giving the adjusted matrix shown in Formula (1).

$$W' = W + \Delta W \tag{4.1}$$

The increment is factorised as $\Delta W = BA$, with $B \in \mathbb{R}^{m \times r}$, $A \in \mathbb{R}^{r \times n}$, and rank $r \ll \min(m, n)$, so only the compact low-rank factors are trained. This factorisation captures the most task-discriminative directions in weight space while drastically reducing the number of trainable parameters. A ReLU non-linearity applied to the intermediate activations introduces further representational depth, after which the second projection maps the expanded d_{ff} representation back to d_{model}, restoring dimensional consistency with the rest of the network. The feedforward sub-layer thereby combines the non-linearly enriched representations with the contextual signals accumulated by the attention layer, strengthening the model's overall expressiveness. Through this architecture, the attention and feedforward sub-layers play complementary roles: prompt-augmented attention layers heighten the model's responsiveness to domain-specific cues, while LoRA-updated feedforward layers amplify representational richness through non-linear low-rank transformations. The combined effect yields strong task adaptation with substantially reduced training cost. Given image I and prompt P, the model generates a diagnostic report $Y = (y_1, y_2, \ldots, y_T)$ via the autoregressive decoding objective in Formula (2).

$$Y = \arg\max \prod_{t=1}^{T} P(y_t \mid y_{<t}, I, P; \theta) \tag{4.2}$$

where θ denotes all model parameters and $y_{<t}$ represents the sequence of tokens generated prior to position t.

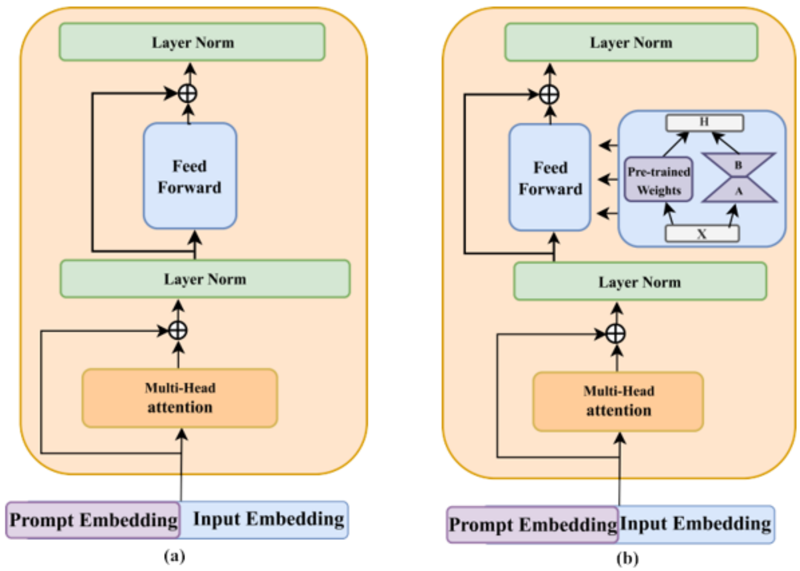


Figure 4.2: Schematic of the P-LoRA adaptation strategy. Panel (a) depicts the soft-prompt insertion applied to intermediate transformer layers; panel (b) shows the selective LoRA injection used in the first, middle, and final layers of the language model backbone.

4.1.4 Data optimization and filtering

The private training corpus for Stage 2 consists of clinical gastroscopy reports obtained from Government Medical College, Calicut. Each record documents a patient's upper gastrointestinal examination, accompanied by endoscopic images spanning multiple anatomical regions: the esophagus, cardia, fundus, body, antrum, pylorus, duodenal bulb, and descending segment. While the images supply the direct visual evidence required for model training, the physician-authored narratives provide the clinical grounding necessary for report generation. In practice, however, a proportion of records contain incomplete entries, non-standard phrasing, or minor transcription errors. Given the well-documented text-processing capabilities of GPT-4 [29,30], we adopt it as an auxiliary annotation tool to refine and enrich these raw sentences. Imposing a uniform template structure across all training descriptions serves two purposes: it eliminates noise introduced by heterogeneous writing styles, and it ensures the model's attention during fine-tuning remains on clinically meaningful content rather than incidental formatting variation. For the templatisation step, GPT-4 receives the following structured prompt: "You are a professional medical text processing assistant. Your task is to convert the following endoscopic image description sentence into a standardized template format: 'This image shows (gastric region). It describes the following features (details of color, texture, presence)'". Quality control is governed by a three-step pipeline: raw text is first submitted to GPT-4 for template conversion; each generated output is then independently reviewed by experienced gastroenterologists, who assess medical accuracy, correct terminology usage, and logical coherence; outputs failing these criteria are revised or removed. Under this workflow, GPT-4 acts strictly as a preprocessing assistant while domain-expert physicians retain final authority over clinical validity. An invariant constraint governing every optimisation step is that the semantics of the revised sentence must be fully preserved; no clinical information may be added, omitted, or distorted in the reformulation. To enforce this requirement, a cosine similarity measure over sentence embeddings is employed. Sentence-BERT [31] encodes both the original and the reformulated sentence into vector representations; denoting these vectors as u and v respectively, the semantic consistency score is given by Formula (3).

$$S = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \tag{4.3}$$

where S denotes the similarity score. A threshold of 0.85 is applied: any reformulated sentence scoring below this value is rejected and regenerated until the criterion is satisfied, ensuring that data enrichment does not compromise the clinical fidelity of the training corpus.

4.1.5 Data Augmentation

Clinical endoscopic records sourced from Government Medical College, Calicut yielded an initial collection of approximately 6,500 physician-annotated reports. Although each record meets high clinical quality standards, this volume alone is inadequate for robust fine-tuning of a large multimodal model. To expand the dataset while preserving clinical validity, two complementary image augmentation strategies are applied: rotation augmentation and stochastic translation augmentation.

Rotation Augmentation

Each original endoscopic image is rotated by 0°, 90°, 180°, and 270°, producing four variants per image. This is clinically safe because tissue textures and lesion appearances are invariant to rotation in endoscopic imaging. The augmentation produces a structured 4× expansion of the base dataset:

- Base reports: 6,500
- After rotation: $6{,}500 \times 4 = 26{,}000$ reports

Each report consists of 960 tokens (784 visual tokens + 176 text tokens), so the rotated corpus amounts to $26{,}000 \times 960 = 24{,}960{,}000$ tokens.

Stochastic Translation Augmentation

To simulate the natural camera jitter that occurs during real endoscopic examinations, we apply stochastic translation to a subset of the rotated images using a Bernoulli selection process with probability $p = 0.2$. For each selected image, the translation vector (τ_x, τ_y) is drawn from a truncated Gaussian distribution:

$$\tau_x \sim \mathcal{N}(0,\, (0.05W)^2), \quad \text{clamped to } [-0.15W,\, +0.15W]$$

$$\tau_y \sim \mathcal{N}(0,\, (0.05H)^2), \quad \text{clamped to } [-0.15H,\, +0.15H]$$

where W and H are the image width and height respectively. This guarantees that at least 72.25% of the original image content is preserved, ensuring that diagnostic features remain intact. The translation augmentation adds approximately $26{,}000 \times 0.2 = 5{,}200$ additional reports, contributing $5{,}200 \times 960 = 4{,}992{,}000$ tokens.

Combined Dataset Size

The combined token count after both augmentations is:

$$24{,}960{,}000 + 4{,}992{,}000 \approx 29{,}952{,}000 \text{ tokens} \approx 29.95\text{M tokens}$$

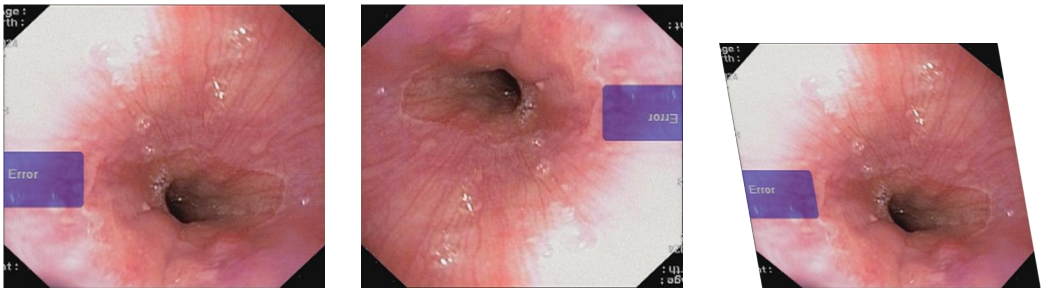
The final second-stage training set thus comprises approximately 31,200 image–report pairs, which substantially expands the original 6,500 clinical reports while preserving clinical validity. A 5% held-out validation split (per the training configuration) is applied to this augmented dataset, yielding approximately 29,640 training samples and 1,560 validation samples.

Figure 4.3 illustrates representative examples of the two augmentation strategies applied to a sample endoscopic image.

4.1.6 Chain-of-Thought Integration

The Interpretability Problem and Motivation

When a model generates diagnostic labels or report text directly from an endoscopic image without exposing its intermediate reasoning, the output is opaque: clinicians have no way to verify or interrogate the inferential steps that led to the conclusion. This "black-box" behaviour is a fundamental obstacle to clinical adoption. To mitigate it, we augment the second-stage training data with chain-of-thought (CoT) reasoning sequences that make the model's decision pathway explicit.



(a) Original (b) Rotated (90°) (c) Stochastic Translation

Figure 4.3: Representative augmentation examples drawn from the GMC Calicut corpus. (a) Original endoscopic frame. (b) Frame after 90° rotation; rotation invariance of gastric lesion morphology makes this transformation clinically safe. (c) Frame after stochastic translation sampled from a truncated Gaussian ($\sigma = 5\%W$, Bernoulli trial $p = 0.2$), preserving at least 72.25% of the original image content.

CoT Implementation

Every training question–answer pair is reformulated to include an explicit step-by-step reasoning chain. The chain imposes a fixed structure on the model's response: *anatomical identification* → *visual observation* → *diagnosis*. Through fine-tuning on these augmented samples, the model learns to predict the complete reasoning chain rather than only the final conclusion, making its inferential process transparent and auditable.

For example, contrast the two modes of response:

Before CoT (baseline):

> *Input:* [Image] "What is the diagnosis?"
> *Output:* "Grade II Esophageal Varices."

After CoT (our model):

> *Input:* [Image] "Analyze the image step-by-step and provide a diagnosis."

Output:
Observation 1: I observe the lower third of the oesophagus.
Observation 2: There are swollen, tortuous submucosal veins present.
Observation 3: The veins occupy less than one third of the lumen, with no red colour signs.
Conclusion: Therefore, the diagnosis is Grade II Esophageal Varices.

Impact on Clinical Utility

Incorporating CoT supervision converts the system from an opaque classifier into an interpretable diagnostic assistant. Three concrete clinical benefits follow: (1) each reasoning step can be verified independently by the reviewing clinician; (2) erroneous intermediate observations become visible and can be corrected before they propagate to the final conclusion; and (3) the structured output format mirrors the conventions of formal clinical documentation, reinforcing medically appropriate language generation. These properties are especially important in real-world deployment settings, where clinical trust and professional accountability are non-negotiable requirements.

Chapter 5

Results

5.1 Experiments and discussion

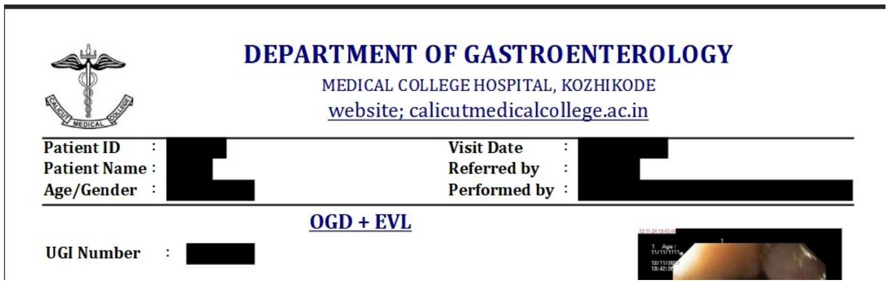
To validate the effectiveness of our proposed method, we conducted fine-tuning on both public and private datasets and evaluated the results across automated metrics and expert physician evaluation.

5.1.1 Datasets

The training corpus is assembled through a two-phase procedure, with the first phase drawing gastrointestinal descriptive content from publicly available medical datasets, as summarised in Table 5.1. A curated gastroenterology terminology list drives automated query-based retrieval, restricting the collected samples to material that is directly relevant to the target clinical domain and excluding records whose subject matter falls outside gastrointestinal medicine. Each candidate report subsequently undergoes an independent quality review by practicing gastroenterologists, who discard samples that are clinically ambiguous, incomplete, or otherwise unsuitable for supervised training, thereby elevating the overall reliability and professional standard of the retained corpus. The underlying sources, namely PubMed

[6], MedTrinity-25M [12], and VQA-Med [6], span a broad range of medical topics and together offer detailed organ-level descriptions alongside extensive disease-specific terminology. After applying the domain-filtering and quality-control steps, the retained subset amounts to approximately 60K records used exclusively for first-phase fine-tuning. The resulting collection furnishes the model with grounded knowledge of gastrointestinal anatomy, characteristic lesion presentations, and domain-appropriate semantic patterns, establishing the representational foundation required for the more specialised second-phase training. The second phase relies on upper gastrointestinal endoscopy reports sourced from Government Medical College, Calicut, which are selected and preprocessed according to the procedure outlined in Section 4.1.4 to yield a tightly focused training set. All records originate from actual clinical encounters and must pass through a purpose-built automated de-identification pipeline before entering the training corpus; this step is mandated by India's Digital Personal Data Protection (DPDP) Act and aligns with HIPAA-equivalent privacy requirements. The pipeline systematically and irreversibly eliminates every Protected Health Information (PHI) field from the text body and its associated metadata: patient identifiers such as name and patient ID are removed, as are age and gender fields, encounter dates, and procedure reference numbers (UGI-OGD report numbers). Because the removal is cryptographically one-way, re-identification from the sanitised records is not feasible. Figure 5.1 illustrates this pipeline as applied to a representative patient report header.

Within the assembled dataset, approximately 40% of cases are classified as normal, with the remaining 60% representing pathological findings; this deliberate imbalance towards abnormal cases ensures the model is exposed to sufficient pathological diversity for discriminative learning. The pathological category covers a broad spectrum of gastrointestinal diagnoses, among them esophageal varices, gastric polyps, gastric bleeding, gastric ulcers, early-stage gastric cancer, and advanced gastric cancer. Coverage of the gastrointestinal



DEPARTMENT OF GASTROENTEROLOGY
MEDICAL COLLEGE HOSPITAL, KOZHIKODE
website; calicutmedicalcollege.ac.in

Patient ID :	[redacted]	Visit Date :	[redacted]
Patient Name :	[redacted]	Referred by :	[redacted]
Age/Gender :	[redacted]	Performed by :	[redacted]

OGD + EVL

UGI Number : [redacted]

Figure 5.1: Automated PHI removal pipeline for GMC Calicut clinical records. Prior to any use in model training, the system irreversibly suppresses all identifying fields, including Patient ID, full name, age and gender, procedure dates, and UGI-OGD reference numbers, to satisfy the requirements of India's Digital Personal Data Protection (DPDP) Act.

tract is systematic: images are drawn from the esophagus, cardia, gastric fundus, body, antrum, and pylorus, together with the duodenal bulb and descending duodenum, guaranteeing both anatomical comprehensiveness and clinical representativeness. All frames are selected intraoperatively by attending endoscopists from diagnostically relevant views during routine examinations. Acquisition hardware varies across endoscopic platforms available at Government Medical College, Calicut, yet a uniform native resolution of 1920×1080 is maintained throughout. For model input, the 1920×1080 captures are downsampled to 336×336 in the preprocessing stage prior to ingestion by the CLIP-ViT-L/14 vision encoder. Training and validation partitions follow a 5% validation split, while a fully independent held-out test set is reserved exclusively for final performance evaluation. Conformity with the standardised acquisition protocol means that all retained images satisfy clinical diagnostic standards with respect to image clarity, anatomical coverage, frame validity, and lesion visibility. To guard against terminological drift or stylistic inconsistency introduced during data optimisation, multiple senior physicians conduct rigorous post-processing reviews of the finalised reports, correcting any deviations in medical accuracy or professional register. Quality verification confirms that all retained records satisfy the predefined similarity criterion, with 78.6% of entries attaining a similarity score above 0.9, a result that reflects a high overall standard of report consistency and reliability. The quality-controlled collection comprises approximately 6,500 physician-annotated endoscopy reports originating from Government Medical College, Calicut. These records are characterised by consistent professional register, reflecting the vocabulary choices, syntactic patterns, and diagnostic conventions that experienced endoscopists employ in formal reporting practice; these attributes are directly transferable to the model's generative output in the target clinical domain. Although the second-stage corpus is substantially smaller than the first-stage public data, its value lies in concentrated clinical specificity: every record is devoted to precise visual description of endoscopic findings and structured documentation of lesion characteristics.

5.1.2 Implementation details

All fine-tuning experiments are built on the LLaVA-4 base model, with P-LoRA applied identically in both training stages. The adapter configuration is held constant across stages: rank = 64, $\alpha = 32$, and dropout = 0.05. Each stage processes data in micro-batches of size 2; four gradient-accumulation steps bring the effective global batch size to 8. A peak learning rate of 2e-4 is adopted together with a warm-up ratio of 0.1, and the token budget per sequence is capped at 4096. Four training epochs are completed under a cosine decay schedule, employing the AdamW optimiser with 8-bit quantisation (adamw_bnb_8bit) and bfloat16 (BF16) mixed precision throughout. Flash Attention 2 is activated to lower the memory cost of self-attention over long contexts. The random seed is fixed at 42 to ensure reproducibility. Greedy decoding (temperature = 0, top_p and top_k both disabled) is used at inference time; the full hyperparameter listing appears in Table 2.

Vision Encoder CLIP-ViT-L/14. Trained on 400M image-text pairs, this encoder delivers reliable medical visual representations; the L-scale

Table 5.1: Dataset composition across both training stages. For each source used in Stage 1 and Stage 2, the table records the origin, data modality, record count, and a brief characterisation of content.

Stage	Data source	Data type	Number	Description
First stage	PubMed; MedTrinity-25M; VQA-Med	Gastrointestinal tract related knowledge	60k	Descriptive data related to the gastrointestinal tract, selected from open medical datasets. Includes fundamental semantic knowledge, lesion descriptions, and anatomical expressions.
Second stage	Physician-annotated data from Government Medical College, Calicut	Specialized endoscopic report data	6.5k (base); ∼31k (after augmentation)	High-quality, professional endoscopic report data, cleaned and optimized for generating template-based language matching physicians' professional style. Expanded via rotation and stochastic translation augmentation.

variant strikes a favourable balance between representational capacity and computational throughput, while the 14×14 patch decomposition simultaneously encodes broad anatomical context and fine-grained lesion detail.

Image resolution 336×336 pixels. This resolution is the native input size for LLaVA-family models and preserves clinically relevant detail at a memory footprint compatible with multi-sample batching.

Patch size 14×14 (24×24 grid).

Total vision tokens 577 per image (576 patch tokens plus one CLS token). The per-patch tokenisation grants the language decoder spatially grounded access to both coarse-grained anatomical layout and localised region-level cues.

Training Precision bfloat16 (BF16). BF16 cuts memory consumption roughly in half relative to FP32 while maintaining numerical stability throughout training.

Attention Flash Attention 2. An exact-attention algorithm that achieves 5–10× VRAM savings for extended token sequences by restructuring memory access patterns.

All experiments are executed on a Linux machine running PyTorch, equipped with dual Intel Xeon 6505P processors (12 cores per socket, 2 sockets, 48 threads in total), 251 GiB of system RAM, and a single NVIDIA H200 NVL GPU (143 GB HBM3e, CUDA 13.0). The software environment consists of Python 3.12.3 and PyTorch 2.11.0+cu130. By applying P-LoRA, only 0.343% of the total model parameters are marked as trainable, leaving the backbone frozen and substantially reducing storage and compute overhead. Peak GPU memory consumption during training reaches approximately 43 GB, and each report is generated in roughly 7 s on average.

Table 5.2: Training hyperparameters for Stage 1 and Stage 2 fine-tuning runs.

Hyperparameter	Stage 1	Stage 2
Learning rate	2e-4	2e-4
Batch size	2	2
Gradient accumulation	4	4
Optimizer	AdamW-8bit	AdamW-8bit
Warm-up ratio	0.1	0.1
Max sequence length	4096	4096
LoRA rank	64	64
α	32	32
Dropout	0.05	0.05
Prompt tokens	64	64
Num epochs	4	4
Random seed	42	42

5.1.3 Evaluation metrics

Following the evaluation protocols established in prior works [32–34], we adopt CIDEr, ROUGE, and METEOR as primary automated metrics to quantify the correspondence between generated reports and ground-truth reference descriptions. CIDEr (Consensus-based Image Description Evaluation) [35] was formulated specifically for image captioning benchmarks. Its distinguishing characteristic is a TF-IDF weighting scheme applied at the n-gram level: n-grams that recur frequently across the corpus are down-weighted as uninformative, whereas rarer, domain-specific phrases receive elevated scores, producing assessments that correlate better with human judgements of caption quality. ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [36] provides an automatic quality estimate for text generation systems and has become standard in summarisation, translation, and captioning evaluation pipelines. The metric quantifies how much content from the reference text is recovered by the generated output, making recall the central quantity of interest. METEOR (Metric for Evaluation of Translation with Explicit Ordering) [37] was developed to remedy known deficiencies in BLEU [38]-class metrics, which inadequately account for morphological variation and paraphrase relations between a candidate and its reference. METEOR combines exact token matching with stem- and synonym-level alignment, yielding sensitivity to both lexical form and semantic equivalence across natural language generation tasks. Because automated metrics capture surface-level textual similarity rather than clinical correctness or diagnostic utility, they alone are insufficient for evaluating medical reports; accordingly, structured expert physician evaluation is incorporated as a complementary assessment.

5.1.4 Experimental results

Comparison of different models

To establish the relative merit of the proposed approach, a set of widely used multimodal large language models, namely Qwen2-VL-7B-Instruct, BLIP-2 (opt-2.7b), and InstructBLIP, is selected as representative baselines, and a comprehensive quantitative comparison is presented in Table 3. Uniform evaluation conditions are maintained by providing each baseline with the identical input prompt used by our model: "Observe this endoscopic image and describe the gastric features presented in the image". Following Zhang et al. [42], the proposed model diverges from the aforementioned baselines by being systematically adapted to the specialised task of endoscopic report generation through targeted fine-tuning of a powerful foundation backbone. The two-stage curriculum (general gastrointestinal knowledge acquisition in the first stage, then professional clinical report generation in the second) allows the model to build domain understanding incrementally rather than attempting to acquire both capacities simultaneously. Complementing this staged schedule, the task-specific architectural design and the domain-tailored fine-tuning protocol together confer markedly improved adaptability to the endoscopic imaging domain. Equally important is the construction of the high-quality, physician-annotated institutional corpus, which provides the precise professional style and clinical grounding that generic datasets cannot supply.

Table 5.3: Quantitative comparison of MVLMERG against baseline vision-language models. Results are reported on the held-out validation split across five automated metrics: CIDEr, ROUGE-L, METEOR, BLEU-4, and BERTScore.

Method	CIDEr ↑	ROUGE-L ↑	METEOR ↑	BLEU-4 ↑	BERTScore ↑
Qwen2-VL-7B-Instruct	0.0078	0.0504	0.0564	0.0009	0.8317
BLIP-2	0.0079	0.0147	0.0347	0.0002	0.7957
InstructBLIP	0.0088	0.0320	0.0583	0.0004	0.8196
Our Model	**0.0531**	**0.2632**	**0.1187**	**0.0006**	**0.8760**

Chapter 6

Conclusion and Future work

6.1 Limitations and future work

For example, outstanding open problems include automated pinpointing of lesion boundaries, reliably grading and classifying gastric malignancies, and accurately interpreting atypical or overlapping lesion presentations. Despite the model's strong overall accuracy in report generation, the outputs are not error-free. Clinically consequential mistakes, such as failure to report early malignancy or active haemorrhage, can directly alter management decisions and therefore represent the most critical failure mode. By contrast, errors that amount to minor phrasing differences or stylistic variation carry little clinical weight. Physician assessment confirmed that the large majority of mistakes belong to the latter, lower-risk category; nonetheless, even rare high-consequence errors underscore why every model-generated report must be reviewed by a qualified clinician before it informs patient care. Taken together, these observations reinforce the principle that AI-assisted tools should be embedded within, rather than substituted for, established clinical review processes. Although the endoscopic equipment at Government Medical College, Calicut covers multiple brands, the same brand may differ across hospitals, and variations in physicians' expertise may also lead to differences in reports. Consequently, training on data drawn from a single institution introduces a risk that performance degrades when the model is applied in other hospital settings. Across institutions, differences in the make and model of endoscopes, the lighting and colour-rendering characteristics of each system, and the varying vocabulary and diagnostic thresholds adopted by individual clinicians can all shift the statistical properties of both images and reports. When the training corpus predominantly reflects the annotation habits of a small cohort of clinicians, the resulting model may inadvertently encode their particular choice of wording, their relative emphasis on certain findings, or their tolerance for diagnostic uncertainty. Furthermore, regional differences in disease prevalence and case complexity may introduce additional bias. These factors can cause domain shifts that reduce the model's performance when applied to unseen data. If endoscopic images present ambiguous or contradictory visual cues, the model is designed to generate conservative, evidence-based descriptions rather than definitive diagnoses. This approach helps prevent over-interpretation and maintains clinical reliability. In addition, automatically flagging instances with high uncertainty for closer physician review represents a highly valuable direction for future work. Future work will involve a concrete multi-center validation plan. Specifically, we plan to collaborate with several hospitals in different regions of India to collect diverse endoscopic image–report pairs from various imaging systems and patient populations. The evaluation will include cross-hospital transfer tests, consistency assessments across different endoscopic brands, and clinical validation by independent physicians. This study primarily focuses on the analysis of individual endoscopic images, where diagnostic conclusions are limited to describing specific lesions observed within each image. However, in real-world clinical practice, endoscopists typically acquire multiple images per patient to form a comprehensive diagnostic assessment. Accordingly, an important direction for future work is to extend the current single-image interpretation framework toward multi-image and patient-level report generation, enabling integrated diagnostic summaries that more closely reflect actual clinical workflows. Building upon this direction, future research will further explore video-based endoscopy analysis by modeling temporal information across consecutive frames. Such temporal modeling has the potential to support more robust lesion localization, progression analysis, and comprehensive patient-level diagnosis. Additionally, regional differences in medical practices and terminology may lead to interpretative biases in the generated medical descriptions. When deploying the proposed model in new hospitals or clinical settings, domain shifts may arise due to differences in endoscopic devices, imaging protocols, patient populations, and report-writing styles. To address this issue, several lightweight adaptation strategies can be adopted. First, parameter-efficient fine-tuning methods can be applied using a small number of locally collected image–report pairs, enabling rapid domain adaptation with limited computational cost. Second, prompt adjustment provides a flexible alternative, where hospital-specific reporting templates, terminology preferences, or anatomical naming conventions can be incorporated into structured prompts without updating model parameters. In addition, continuous feedback from clinicians can be used to iteratively refine prompts or select representative samples for incremental fine-tuning. These strategies allow the model to be efficiently adapted to new hospitals while preserving data privacy and minimizing deployment overhead. Regarding the application of large language models in healthcare, the foremost challenge lies in data availability. Collecting endoscopic data is time-consuming and costly, so it is important to explore how LLM-based approaches can generate synthetic data that closely reflect real clinical scenarios while adhering to ethical standards [40,41]. Furthermore, in terms of LLM architecture, an important future direction is to design models that effectively leverage the strengths of both language and visual encoders, achieving efficient multimodal feature fusion. Research on clinical endoscopy videos also holds greater practical significance and applicability. Finally, direct inference with LLMs may not fully exploit their potential; existing strategies such as rethinking or chain-of-thought reasoning may enhance their reasoning capabilities, representing another highly valuable direction for future work. We are actively communicating with Government Medical College, Calicut, to seek approval for two concurrent goals: expanding the dataset with additional physician-annotated endoscopic cases and releasing a de-identified subset for the broader research community. A larger and more diverse clinical dataset would directly improve model generalizability, reduce domain bias, and enable finer-grained classification of gastric and oesophageal conditions. Both data expansion and public release are subject to institutional review and privacy regulations under India's Digital Personal Data Protection (DPDP) Act, and we are committed to making as much data as ethically permissible accessible to support further research in automated endoscopic report generation. In addition, the pretrained model, P-LoRA configurations, and training scripts will be made available upon publication to support reproducibility and further research. To enable practical clinical deployment, we plan to explore model quantization strategies to reduce memory footprint while preserving inference quality. A RESTful API layer and electronic health record (EHR) integration module are planned to facilitate seamless adoption into existing clinical workflows. Full clinical deployment is contingent on formal approval from the Institutional Ethics Committee (IEC) of Government Medical College, Calicut, and compliance with applicable regulatory standards.

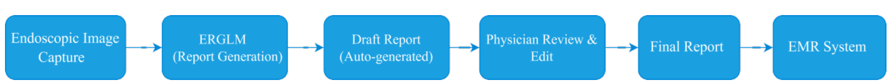


Figure 6.1: Intended deployment workflow for MVLMERG in a clinical setting: an endoscopic image is ingested by the system, which produces a structured draft report; a reviewing clinician edits and approves the draft before it is committed to the electronic medical record (EMR).

6.2 Conclusion

Building on the P-LoRA framework introduced by Zhang et al. [42], this study constructs an endoscopic image interpretation system that combines prompt-conditioned low-rank adaptation with a two-stage fine-tuning curriculum, designed to unlock the latent diagnostic capacity of pre-trained multimodal foundation models. Following both fine-tuning stages, the resulting model produces clinically detailed endoscopic descriptions and demonstrates notably strong sensitivity to morphological image characteristics. Across all quantitative benchmarks, the proposed system achieves substantially higher scores than the evaluated comparative models, including GPT-4V and HuatuoGPT-Vision-7B.

CRediT authorship contribution statement

Praval Pattam: Software, Writing – original draft, Methodology, Conceptualization. Theenesh Potluri: Software, Writing – review & editing, Validation, Formal analysis. Pranav Sai Sarvepalli: Software, Data curation, Investigation.

Ethics and safety

All procedures in this study comply with ethical standards for medical data usage. The dataset used for model training and evaluation is fully de-identified by the data-providing institution, with all Protected Health Information (PHI) removed in accordance with HIPAA-like privacy guidelines. No patient-identifiable information is accessed or processed during the development of this study. To ensure safe deployment, all model-generated outputs are intended solely for research purposes. The system is not designed for direct clinical application, and any generated report is required to undergo review and verification by licensed physicians before being used in clinical decision-making. In addition, all experiments are approved by the corresponding institutional ethics committee, and data usage strictly follows the institution's data governance requirements.

Ethics statements

The research described in this paper was carried out in conformity with the ethical requirements of Government Medical College, Calicut, and the National Institute of Technology Calicut. All datasets employed in this work were treated with full confidentiality and restricted to academic research use.

Declaration of competing interest

The authors confirm that no financial interests, personal relationships, or affiliations exist that could be perceived as influencing the outcomes or conclusions presented in this work.

Data availability

The data underlying this study cannot be shared due to institutional restrictions and patient privacy obligations.

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